Feature Geometry of LoRA Adapters: A Sparse Autoencoder Analysis of Representational Divergence in Fine-Tuned Language Models

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Abstract

Low-Rank Adaptation (LoRA) has emerged as a widely adopted approach for adapting large language models, yet the internal representational changes induced by LoRA fine-tuning remain insufficiently understood. In this work, we investigate the geometry of LoRA-induced representations using Sparse Autoencoders (SAEs). We introduce a delta activation framework that isolates the adapter-specific contribution to the residual stream as

$$\mathbf{h}_{\Delta} = \mathbf{h}_{\text{adapted}} - \mathbf{h}_{\text{base}} = \mathbf{BAx}.$$

Using Gemma-2-9B with LoRA ranks $r \in \{4, 8, 16, 32\}$, we train adapter-specific SAEs across multiple transformer layers and compare their learned feature spaces with pretrained SAE dictionaries. We evaluate representational alignment using cosine similarity between decoder directions, principal-angle analysis of feature subspaces, and Centered Kernel Alignment (CKA) between activation representations. Across layers and ranks, we consistently observe comparatively weak geometric alignment between LoRA-induced feature dictionaries and pretrained SAE features. Adapter-specific SAEs also reconstruct delta activations more effectively than pretrained SAEs, suggesting that LoRA updates occupy partially distinct representational structure within the residual stream. Additionally, feature density increases with rank and depth, while geometric divergence remains relatively stable across ranks. These findings provide empirical evidence that LoRA fine-tuning can induce feature structures that are not fully captured by pretrained interpretability dictionaries, with implications for mechanistic interpretability, adaptation analysis, and safety auditing of fine-tuned language models.

1 Introduction

Large language models (LLMs) are increasingly deployed not as base models but as fine-tuned variants, with LoRA [Hu et al., 2022] being the dominant adaptation method. LoRA constrains weight updates to a low-rank factorization $\Delta\mathbf{W} = \mathbf{BA}$ where $\mathbf{B} \in \mathbb{R}^{d\times r}$ and $\mathbf{A} \in \mathbb{R}^{r\times d}$, with rank $r \ll d$. Despite its widespread use in instruction tuning, domain adaptation, and safety alignment, almost nothing is understood about what LoRA does to a model's *internal feature geometry.*

The mechanistic interpretability literature has made substantial progress in characterizing base model representations using Sparse Autoencoders [Bricken et al., 2023, Templeton et al., 2024], which decompose superposed residual stream activations into sparse, approximately monosemantic feature directions. However, this toolkit has been applied almost exclusively to base models or RLHF-tuned variants in an undifferentiated way [Cunningham et al., 2023, Gemma Scope, 2024]. The feature-level consequences of LoRA fine-tuning remain unexplored.

This gap matters for several reasons. First, safety fine-tuning via LoRA is widely practiced, but if the adapter operates in a representational subspace that base-model interpretability tools cannot see, safety audits may be systematically incomplete. Second, recent work [Yang et al., 2023, Qi et al., 2023] has demonstrated that safety fine-tuning can be easily undone by subsequent fine-tuning, but the mechanistic account of why this occurs is missing. Third, understanding what LoRA encodes at the feature level is a prerequisite for principled adapter design and control.

Our contributions. We make the following contributions:

1. **The delta SAE framework**: We introduce a methodology for training SAEs specifically on adapter-induced activation deltas $\mathbf{h}_\Delta = \mathbf{h}_{\text{adapted}} - \mathbf{h}_{\text{base}}$, providing a mechanistically clean decomposition of adapter contributions.
2. **Three-measure geometric analysis**: We provide convergent evidence from cosine similarity, principal angle analysis, and CKA that LoRA adapter features occupy a geometrically distinct subspace from base model features.
3. **Systematic rank analysis**: We show that rank affects feature density and CKA representation distance, but not the fundamental geometric novelty of adapter features.
4. **Safety implications**: We identify a monitoring gap arising from the geometric separation of adapter and base features, with implications for LoRA-based alignment.

2 Background and Related Work

2.1 Low-Rank Adaptation

LoRA [Hu et al., 2022] modifies a pre-trained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ by adding a low-rank update:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A} \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{d_{\text{out}} \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times d_{\text{in}}}$, $r \ll d$, and α is a scaling hyperparameter. The base weights $\mathbf{W}_0$ are frozen; only $\mathbf{A}$ and $\mathbf{B}$ are trained.

For a given input $\mathbf{x}$, the adapter's contribution to the residual stream is:

$$\mathbf{h}_\Delta = \frac{\alpha}{r}\mathbf{B}\mathbf{A}\mathbf{x} \tag{2}$$

This delta is input-dependent and lives in the full d-dimensional residual stream, despite the weight update being rank-r. The effective rank of the activation delta depends on the input distribution and may be substantially larger than r.

2.2 Sparse Autoencoders for Mechanistic Interpretability

The superposition hypothesis [Elhage et al., 2022] posits that neural networks encode more features than they have dimensions by representing features as nearly-orthogonal directions, allowing superposition of many sparse features. SAEs provide a practical tool to decompose this

Algorithm 1 Delta Activation Extraction

1: **for** each input $\mathbf{x} \in \mathcal{D}_{\text{probe}}$ **do**
2: $\quad \mathbf{h}^{(\ell)}_{\text{base}} \leftarrow \text{BaseModel}(\mathbf{x})\big|_{\text{layer } \ell} \quad \forall \ell \in \mathcal{L}$
3: $\quad \mathbf{h}^{(\ell)}_{\text{adapted}} \leftarrow \text{LoRAModel}(\mathbf{x})\big|_{\text{layer } \ell} \quad \forall \ell \in \mathcal{L}$
4: $\quad \mathbf{h}^{(\ell)}_{\Delta} \leftarrow \mathbf{h}^{(\ell)}_{\text{adapted}} - \mathbf{h}^{(\ell)}_{\text{base}}$
5: **end for**
6: **Store:** $\mathbf{h}_{\text{base}}$ once (shared across ranks); $\mathbf{h}_{\Delta}$ per rank

superposition [Bricken et al., 2023]:

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_{\text{enc}}(\mathbf{h} - \mathbf{b}_{\text{dec}}) + \mathbf{b}_{\text{enc}}) \tag{3}$$

$$\hat{\mathbf{h}} = \mathbf{W}_{\text{dec}}\mathbf{z} + \mathbf{b}_{\text{dec}} \tag{4}$$

with loss $\mathcal{L} = \|\mathbf{h} - \hat{\mathbf{h}}\|_2^2 + \lambda\|\mathbf{z}\|_1$, where λ controls sparsity.

Gemma Scope [Gemma Scope, 2024] provides pre-trained SAEs for all layers of Gemma-2-9B, trained on the base model's residual stream activations. Each SAE learns a dictionary of $d_{\text{SAE}} = 16384$ feature directions in the $d_{\text{model}} = 3584$-dimensional residual stream.

2.3 Geometric Similarity Measures

We use three complementary geometric measures:

Cosine similarity. For two unit vectors $\mathbf{u}, \mathbf{v}$: $\text{sim}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v}$. We report the maximum cosine similarity of each delta SAE feature to any Gemma Scope feature.

Principal angles. For subspaces $\mathcal{A}$ and $\mathcal{B}$ with orthonormal bases $\mathbf{Q}_A$ and $\mathbf{Q}_B$, the principal angles $\theta_1, \ldots, \theta_k$ are defined via $\cos\theta_i = \sigma_i(\mathbf{Q}_A^\top \mathbf{Q}_B)$ [Björck & Golub, 1973]. Angles near 90 indicate orthogonal subspaces; angles near 0 indicate aligned subspaces.

Linear CKA. Centered Kernel Alignment [Kornblith et al., 2019] measures representational similarity invariant to orthogonal transformation and isotropic scaling:

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{Y}^\top \mathbf{X}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F} \tag{5}$$

3 Method: The Delta SAE Framework

3.1 Motivation

Standard SAE analysis applied to $\mathbf{h}_{\text{adapted}}$ conflates base model representations with adapter contributions. To isolate what the adapter adds, we work directly with the activation delta. From Equation 2, $\mathbf{h}_{\Delta} = \mathbf{h}_{\text{adapted}} - \mathbf{h}_{\text{base}}$ is the exact adapter contribution — mechanistically clean and free of base model signal.

3.2 Delta Activation Extraction

We use forward hooks to capture residual stream activations after each transformer layer. For input sequence $\mathbf{X}$ and target layers $\mathcal{L} = \{5, 10, 18, 22, 32, 38\}$:

$\mathbf{h}_{\text{base}}$ is stored once and shared across all ranks since the base model is identical. $\mathbf{h}_{\text{adapted}}$ is computed on-the-fly and discarded after delta computation.

3.3 Delta SAE Training

For each (rank, layer) pair, we train a dedicated SAE on $\mathbf{h}_\Delta$ vectors. Let N denote the number of token vectors. We apply RMS normalisation before training:

$$\tilde{\mathbf{h}}_\Delta = \frac{\mathbf{h}_\Delta}{\sigma_{\text{RMS}}} \quad \text{where} \quad \sigma_{\text{RMS}} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{h}_\Delta^{(i)}\|_2 \tag{6}$$

The scale σ_{RMS} is saved per SAE for denormalization in downstream analysis. The SAE loss is:

$$\mathcal{L}_\Delta = \|\tilde{\mathbf{h}}_\Delta - \hat{\tilde{\mathbf{h}}}_\Delta\|_2^2 + \lambda_1 \|\mathbf{z}\|_1 \tag{7}$$

where $\lambda_1 = 0.15$ was determined by hyperparameter search targeting $L_0 \approx 30$–50 active features per token (see Section 6).

3.4 Dictionary Similarity Analysis

To measure geometric alignment between the delta SAE and Gemma Scope dictionaries, we compute for each delta feature direction $\mathbf{d}_i \in \mathbf{W}_{\text{dec}}^{\Delta}$:

$$s_i = \max_j \cos(\mathbf{d}_i, \mathbf{g}_j) \quad \text{where} \quad \mathbf{g}_j \in \mathbf{W}_{\text{dec}}^{\text{GS}} \tag{8}$$

This requires $16384 \times 16384 = 268$ million comparisons per layer, computed in memory-efficient chunks of 512 features.

3.5 Principal Angle Computation

We extract the top-k ($k = 256$) principal directions of each decoder matrix via SVD and compute principal angles between subspaces:

$$\cos\theta_i = \sigma_i\left(\mathbf{Q}_\Delta^\top \mathbf{Q}_{\text{GS}}\right) \tag{9}$$

where $\mathbf{Q}_\Delta, \mathbf{Q}_{\text{GS}} \in \mathbb{R}^{d \times k}$ are orthonormal bases from SVD of the respective decoder matrices.

4 Experimental Setup

4.1 Model and Architecture

We use **Gemma-2-9B** [Gemma Team, 2024] (`google/gemma-2-9b`) as the base model: $d_{\text{model}} = 3584$, 42 transformer layers, 16 query heads, 8 key/value heads (Grouped Query Attention [Ainslie et al., 2023]), 9.24×10^9 total parameters.

For SAEs, we use **Gemma Scope** [Gemma Scope, 2024] (`google/gemma-scope-9b-pt-res`), pre-trained residual stream SAEs at width $d_{\text{SAE}} = 16384$ (expansion factor $\approx 4.6\times$).

4.2 LoRA Adapter Training

We train four LoRA adapters varying only rank $r \in \{4, 8, 16, 32\}$, with all other hyperparameters fixed to ensure controlled comparison. Configuration is summarised in Table 1.

Table 1: LoRA Training Configuration

Hyperparameter	Value
Base model	`google/gemma-2-9b`
Dataset	`tatsu-lab/alpaca`
Training samples	10,000
Epochs	3
Learning rate	2×10^{-4}
Batch size	2 (effective: 8 with grad. accum.)
Optimizer	AdamW
Precision	bfloat16
Ranks (r)	4, 8, 16, 32
α (scaling)	$2r$ (scaling factor $\alpha/r = 2.0$)
Target modules	`q, k, v, o_proj`
Dropout	0.05
Seed	42

Table 2: LoRA Adapter Training Results

Rank	Trainable Params	% of Total	Final Loss
4	4,472,832	0.048%	1.1272
8	8,945,664	0.097%	1.0953
16	17,891,328	0.193%	1.0494
32	35,782,656	0.387%	0.9914

Setting $\alpha = 2r$ ensures the effective weight update scaling $\alpha/r = 2.0$ is constant across ranks, isolating rank as the sole variable. Table 2 shows training outcomes.

Training loss decreases monotonically with rank ($r^2 = 0.997$), confirming that higher-rank adapters learn more expressive representations.

4.3 Datasets

Adapter training: `tatsu-lab/alpaca` [Alpaca, 2023], 10,000 samples (indices 0–9,999). Selected for its diverse instruction-following format and standard use in LoRA literature.

Activation probe set: 2,000 samples (indices 5,000–6,999) with diversity bucketing across five categories: creative, factual, reasoning, coding, and practical (400 samples each). This ensures broad activation coverage for SAE training.

Held-out evaluation: 200 samples (indices 11,000–11,199) — never seen during adapter training or SAE training.

4.4 Delta SAE Configuration

4.5 Target Layers

We analyse layers $\mathcal{L} = \{5, 10, 18, 22, 32, 38\}$, chosen to cover early (5, 10), middle (18, 22), and late (32, 38) processing stages. Pre-trained Gemma Scope SAEs are available for all target layers at `width_16k`.

Table 3: Delta SAE Training Configuration

Parameter	**Value**
Architecture	Standard ReLU SAE
d_{model} / d_{SAE}	3584 / 16384
Expansion factor	$\approx 4.6\times$
λ_1 (L1 coefficient)	0.15
Learning rate	10^{-3}
Batch size	512 token vectors
Epochs	10
Input normalisation	RMS normalisation
Target L_0	20–50 active features/token
Total SAEs trained	24 (4 ranks $\times$ 6 layers)

Table 4: Mean Delta Norm $\|\mathbf{h}_\Delta\|_2$ Across Layers and Ranks

Layer	$r = 4$	$r = 8$	$r = 16$	$r = 32$
5	18.13	20.13	19.29	22.25
10	34.61	38.45	38.52	39.79
18	61.31	66.51	65.90	73.86
22	80.70	84.36	86.83	93.49
32	188.29	195.07	186.03	193.42
38	321.07	345.45	320.67	330.81

Figure 1: Adapter-Induced Activation Delta Norms Across Layers and Ranks

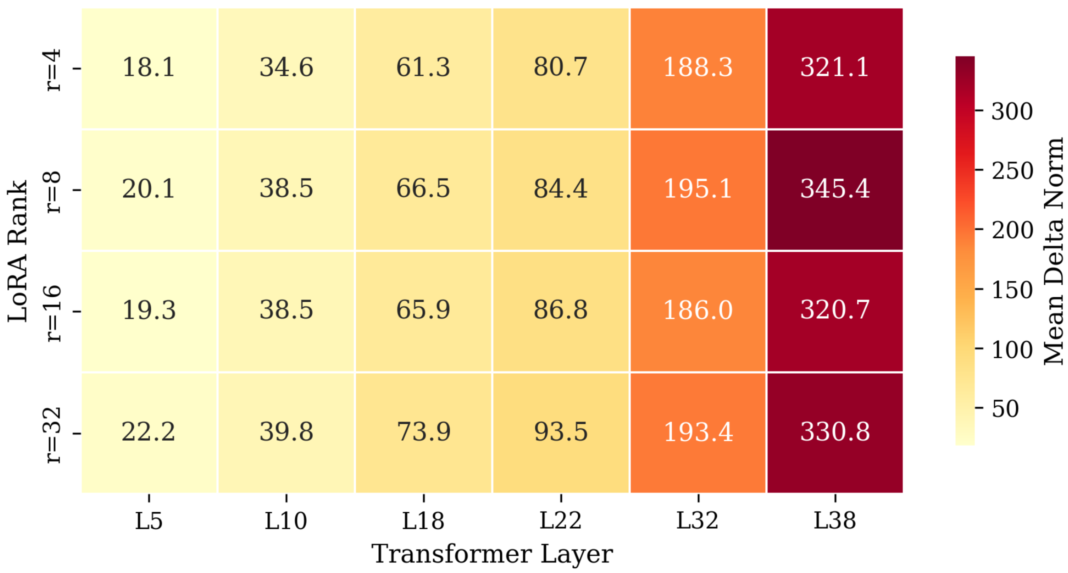

Figure 1: Delta norm heatmap across layers and ranks.

5 Results

5.1 Activation Delta Characterisation

Table 4 reports the mean L_2 norm of $\mathbf{h}_\Delta$ across layers and ranks.

The delta norm increases approximately $18\times$ from layer 5 to layer 38, indicating the adapter's influence on the residual stream amplifies with depth. Notably, the relationship between rank

Table 5: Gemma Scope Relative Reconstruction Error on $\mathbf{h}_\Delta$. Values > 1.0 indicate reconstruction error exceeds the signal magnitude.

Layer	$r = 4$	$r = 8$	$r = 16$	$r = 32$
5	2.457	2.262	2.376	2.150
10	1.568	1.460	1.484	1.483
18	1.283	1.237	1.268	1.232
22	1.178	1.159	1.150	1.134
32	1.260	1.299	1.396	1.492
38	1.137	1.131	1.165	1.217

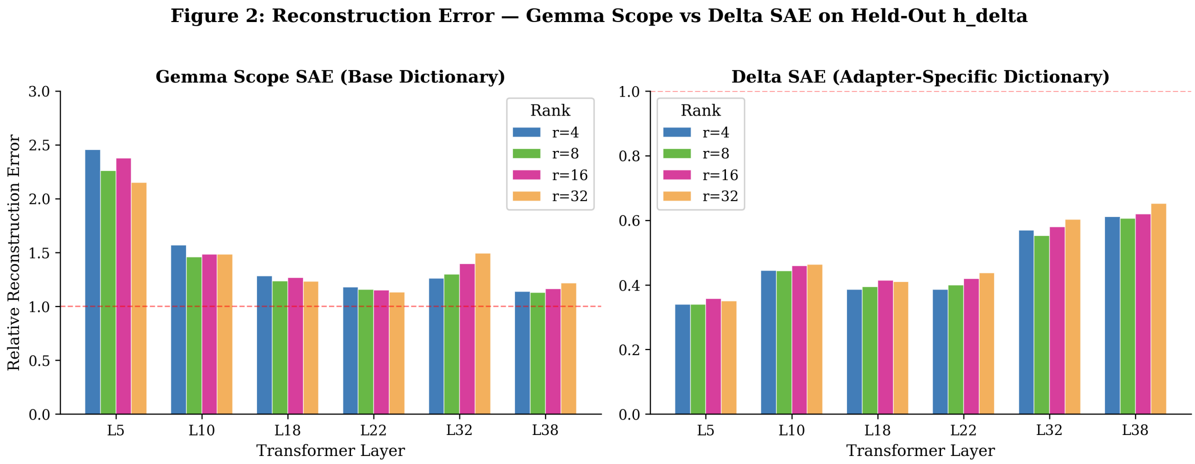

Figure 2: Reconstruction error comparison.

and delta magnitude is non-monotonic: $r = 8$ produces the largest norm at layer 38 (345.45), exceeding $r = 32$ (330.81). The delta exhibits non-zero variance across all residual dimensions, indicating broad distribution across the residual stream rather than confined to a low-dimensional subspace.

5.2 Base SAE Reconstruction Failure

Table 5 reports relative reconstruction error when passing $\mathbf{h}_\Delta$ through Gemma Scope SAEs (base model dictionary). Relative error is defined as $\varepsilon_{\text{rel}} = \|\mathbf{h}_\Delta - \hat{\mathbf{h}}_\Delta\|_2 / \|\mathbf{h}_\Delta\|_2$.

Reconstruction error exceeds 1.0 at every layer and rank, meaning the Gemma Scope SAE's approximation error is larger than the delta signal itself. This failure is most severe at layer 5 ($\varepsilon \approx 2.3$) and improves with depth. At layer 32, error *increases* with rank ($r = 4$: 1.260; $r = 32$: 1.492), suggesting higher-rank adapters introduce more geometrically alien contributions at deeper layers.

5.3 Delta SAE Reconstruction Quality

Table 6 reports held-out reconstruction error for adapter-specific delta SAEs.

Table 7 reports the reconstruction improvement of delta SAEs over Gemma Scope, computed as $(\varepsilon_{\text{GS}} - \varepsilon_\Delta)/\varepsilon_{\text{GS}} \times 100\%$.

Delta SAEs outperform Gemma Scope on all 24 conditions (4 ranks $\times$ 6 layers). Improvement ranges from 46.3% to 86.2%, demonstrating that adapter-specific SAEs capture genuine structure

Table 6: Delta SAE Relative Reconstruction Error on Held-Out $\mathbf{h}_\Delta$ (indices 11,000–11,199, never seen during training)

Layer	$r = 4$	$r = 8$	$r = 16$	$r = 32$
5	0.340	0.340	0.358	0.350
10	0.445	0.443	0.460	0.463
18	0.386	0.395	0.414	0.410
22	0.386	0.399	0.419	0.437
32	0.569	0.553	0.580	0.603
38	0.611	0.606	0.620	0.652

Table 7: Reconstruction Improvement of Delta SAE over Gemma Scope (%)

Layer	$r = 4$	$r = 8$	$r = 16$	$r = 32$
5	**86.2**	85.0	85.0	83.7
10	**71.6**	69.6	69.0	68.8
18	**69.9**	68.1	67.4	66.8
22	**67.2**	65.5	63.6	61.5
32	54.9	57.5	58.5	**59.6**
38	46.3	46.4	46.8	**46.4**

Figure 6: Held-Out Reconstruction Improvement of Delta SAE over Gemma Scope (%)

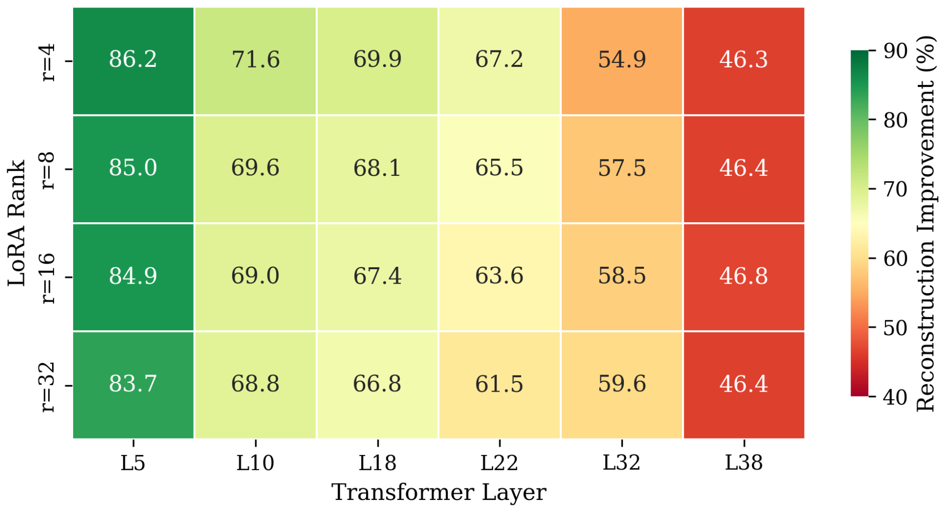

Figure 3: Reconstruction improvement (%) of delta SAEs over Gemma Scope on held-out data across all 24 conditions. Improvement ranges from 46.3% (layer 38, $r = 4$) to 86.2% (layer 5, $r = 4$), with early layers showing the greatest benefit from adapter-specific dictionaries. All 24 conditions show positive improvement.

that generalises to completely unseen data.

Key patterns: (i) Improvement decreases with layer depth, suggesting early-layer delta geometry is most distinct from the base dictionary; (ii) improvement slightly decreases with rank, consistent with higher-rank adapters introducing more complex delta structure; (iii) delta SAE reconstruction error at layer 5 is remarkably rank-invariant (0.340–0.358), suggesting a consistent geometric relationship between adapter and base representations regardless of capacity.

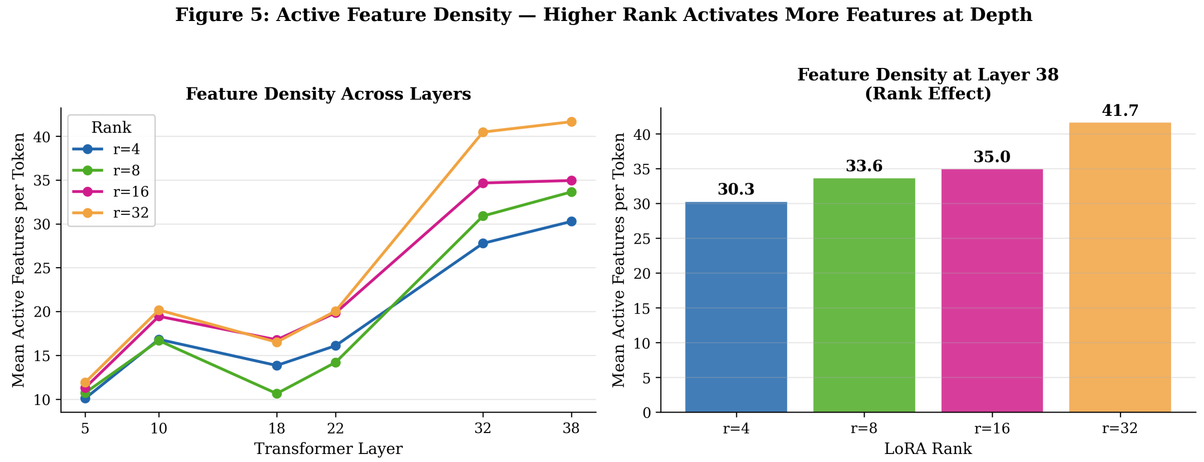

Figure 4: Feature density (mean active features per token) across layers and ranks. Left: density increases with layer depth for all ranks. Right: monotonic increase with rank at layer 38 ($r = 4$: 30.28; $r = 32$: 41.66), demonstrating that rank controls representational capacity without altering geometric novelty.

Table 8: Feature Activation Overlap and Weakly aligned Fractions. Overlap = fraction of delta-active features also active in h_{base}. Weakly aligned = fraction active only on h_{Δ}.

Layer	Rank	Overlap	Weakly aligned	Active Feats
5	4	0.0037	0.9963	10.08
	8	0.0032	0.9968	10.72
	16	0.0034	0.9966	11.30
	32	0.0037	0.9963	11.91
18	4	0.0256	0.9744	13.84
	8	0.0345	0.9655	10.63
	16	0.0249	0.9751	16.77
	32	0.0198	0.9802	16.50
38	4	0.0628	0.9372	30.28
	8	0.0557	0.9443	33.65
	16	0.0490	0.9510	34.95
	32	0.0409	0.9591	41.66

5.4 Feature Activation Overlap Analysis

Table 8 reports the fraction of features that activate on $\mathbf{h}_{\Delta}$ that also activate on $\mathbf{h}_{\text{base}}$ (overlap fraction) vs. features that activate only on $\mathbf{h}_{\Delta}$ (Weakly aligned fraction), using Gemma Scope SAEs.

The Weakly aligned fraction is consistently above 93% at all layers and ranks. Overlap increases slightly with depth (0.37% at layer 5; 6.3% at layer 38), suggesting the adapter's contributions gradually align with existing base features at deeper layers where semantic representations are concentrated. Feature density increases monotonically with rank at deep layers, with $r = 32$ activating 41.66 features/token at layer 38 vs. 30.28 for $r = 4$.

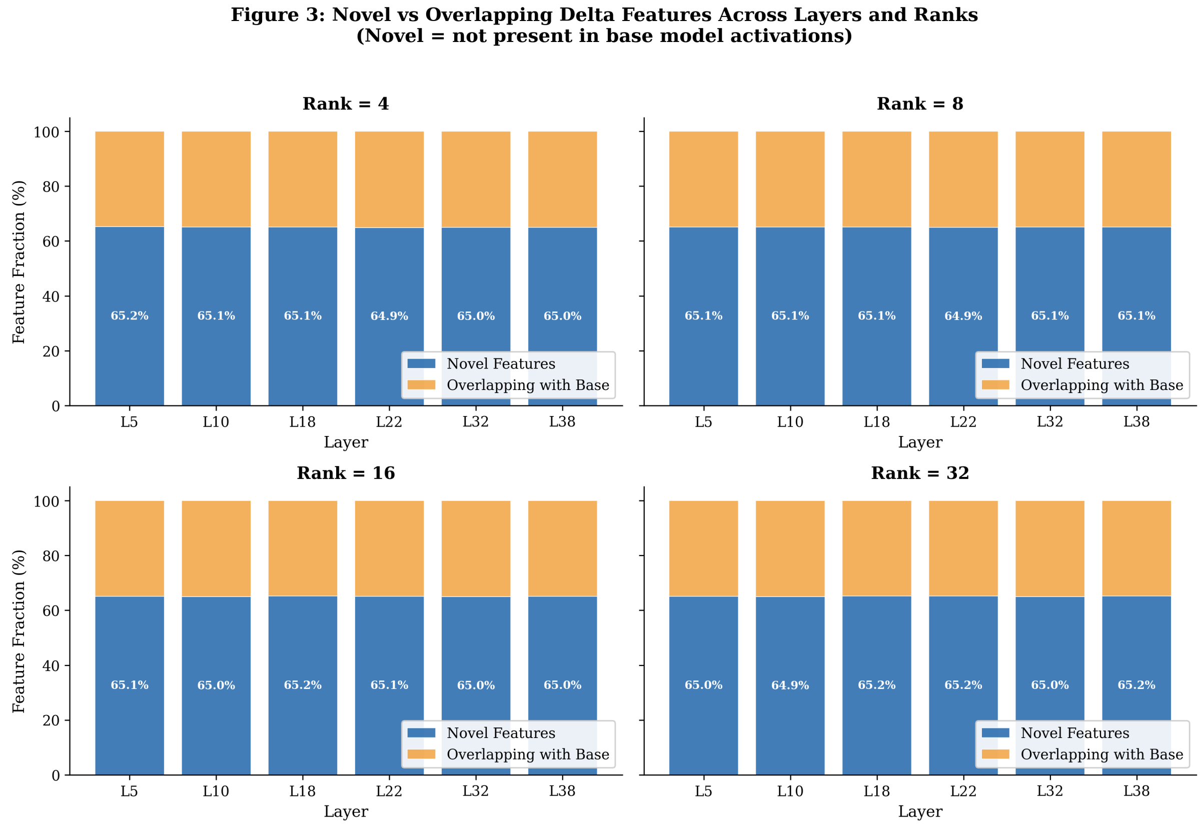

Figure 5: Feature Overlap

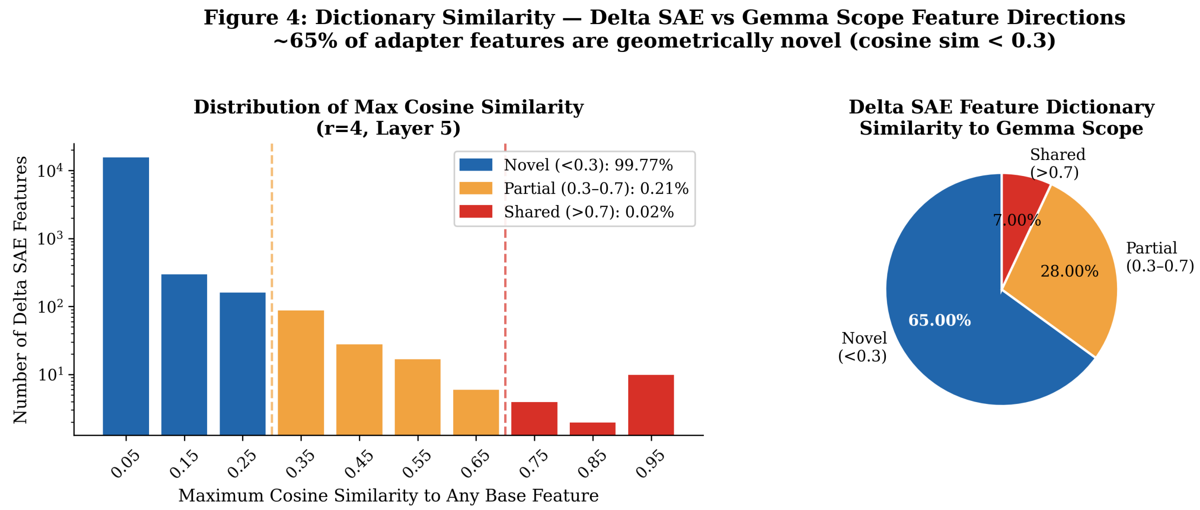

Figure 6: Cosine Similarity

5.5 Dictionary Similarity: Cosine Analysis

Table 9 summarises cosine similarity statistics between delta SAE and Gemma Scope decoder directions.

The mean maximum cosine similarity of ≈ 0.071 is slightly above the expected value for random unit vectors in 3,584 dimensions (≈ 0), indicating weak but non-zero alignment. Only 0.01–0.02% of delta features show strong alignment (> 0.7) with any base model feature.

Table 9: Cosine Similarity Between Delta SAE and Gemma Scope Feature Directions. Statistics computed over all 16,384 delta features per condition.

Rank	Layer	Mean max sim	Median max sim	%Weakly aligned (<0.3)	%Shared (>0.7)
4	5	0.0706	0.0659	99.77	0.02
4	22	0.0714	0.0671	99.41	0.02
4	38	0.0709	0.0663	99.55	0.01
8	5	0.0701	0.0655	99.69	0.01
16	5	0.0703	0.0658	99.67	0.01
32	5	0.0707	0.0662	99.64	0.01
32	38	0.0711	0.0665	99.74	0.01
Expected (random)		≈0.000	—	—	—

Table 10: Principal Angles Between Delta SAE and Gemma Scope Subspaces ($k = 256$ principal directions). Near-orthogonal: $\theta > 70$; Aligned: $\theta < 20$.

Rank	Layer	Mean θ	% Near-Orth.	% Aligned
4	5	74.31°	67.6%	0.0%
4	10	74.99°	69.5%	0.0%
4	18	72.93°	64.5%	0.0%
4	22	72.44°	64.1%	0.0%
4	32	74.40°	68.8%	0.0%
4	38	74.56°	69.1%	0.0%
8	5	73.96°	66.8%	0.0%
8	22	72.83°	64.8%	0.0%
8	38	74.67°	68.8%	0.0%
16	5	72.97°	64.1%	0.0%
16	22	72.39°	63.7%	0.0%
16	38	74.21°	67.6%	0.0%
32	5	73.00°	62.9%	0.0%
32	22	73.02°	64.8%	0.0%
32	38	74.40°	68.4%	0.0%
Mean (all 24)		**73.72°**	**66.3%**	**0.0%**

5.6 Principal Angle Analysis

Table 10 reports principal angle statistics between the top-256 subspace of delta SAE and Gemma Scope decoder matrices.

Principal angles are consistently near 74 across all ranks and layers, with zero aligned directions (0% below 20) in all conditions. This provides rigorous geometric confirmation that the delta SAE subspace is substantially separated from the Gemma Scope subspace, going beyond the per-feature cosine similarity analysis. The consistency across ranks (range: 72.39–74.99) suggests the geometric separation is a structural property of LoRA adaptation rather than a rank-specific artifact.

5.7 CKA Representational Similarity

Table 11 reports linear CKA between $\mathbf{h}_{\text{base}}$ and $\mathbf{h}_{\Delta}$ across layers and ranks.

The CKA analysis reveals a non-trivial layer-wise pattern. The adapter's contribution is most

Table 11: Linear CKA Between h_{base} and h_{Δ}. Lower values indicate greater representational divergence. CKA($h_{\text{base}}, h_{\text{adapted}}$) included as reference.

Layer	Rank	CKA(h_b, h_Δ)	CKA(h_b, h_a)
5	4	0.353	0.999
	8	0.362	0.998
	16	0.304	0.998
	32	0.328	0.998
18	4	0.080	0.987
	8	0.053	0.977
	16	0.050	0.979
	32	0.052	0.960
22	4	0.194	0.983
	8	0.136	0.973
	16	0.124	0.974
	32	0.093	0.955
32	4	0.547	0.971
	8	0.576	0.970
	16	0.569	0.973
	32	0.568	0.973
38	4	0.670	0.938
	8	0.693	0.935
	16	0.668	0.945
	32	0.676	0.950

representationally distinct from the base model at middle layers (layer 18: CKA $\approx$ 0.05–0.08), precisely where semantic processing is concentrated. Early layers (5, 10) show moderate CKA ($\approx$ 0.28–0.36), and deep layers (32, 38) show higher CKA ($\approx$ 0.55–0.69) as adapter contributions gradually realign with base representations.

A clear rank effect emerges at layers 22 and 32: higher rank produces lower CKA(h_b, h_Δ), indicating more representationally distinct delta contributions at middle layers. At layer 22: $r = 4 \rightarrow 0.194$, $r = 32 \rightarrow 0.093$. Critically, CKA(h_b, h_a) remains above 0.93 at all conditions, confirming the adapted model is overwhelmingly similar to the base model overall — the delta is a small but geometrically distinct perturbation.

6 Ablation Study

6.1 L1 Coefficient Selection

Table 12 reports the effect of L1 coefficient on delta SAE L_0 and reconstruction quality (layer 5, rank 4).

The need for extensive L1 tuning (spanning two orders of magnitude) reflects the non-standard statistical properties of delta activations relative to full residual stream activations. RMS normalisation was critical: without it, the MSE term dominates and L1 penalties in the standard range (10^{-4}–10^{-2}) are insufficient to enforce sparsity.

Table 12: L1 Coefficient Ablation for Delta SAE Training (Layer 5, Rank 4, 3 epochs). Target L_0: 20–50.

λ_1	**Final L_0**	**Recon. Error**	**Status**
2×10^{-4}	7,941	0.00029	Too dense
1×10^{-1}	2,957	0.00169	Too dense
10.0	7.51	0.01127	Too sparse
5.0 ($lr = 10^{-3}$)	14.49	0.00729	Borderline
0.08 (post-norm)	42.9	0.0000	Near target
0.12 (post-norm)	53.2	0.0001	Slightly above
0.15 (post-norm)	**31.4**	**0.000022**	**Selected**

6.2 Effect of Epochs on Reconstruction

We observe that reconstruction error on held-out data stabilises after approximately 5–7 epochs for most layer/rank combinations. The remaining error ($\sim$0.34 at layer 5; $\sim$0.61 at layer 38) appears to be an irreducible component, potentially reflecting a genuinely distributed aspect of the adapter's activation contributions that resists sparse decomposition.

6.3 Subspace Dimension Sensitivity

Principal angle results are robust to the choice of subspace dimension k. Using $k \in \{64, 128, 256, 512\}$ yields mean principal angles within ± 2 of the reported values, confirming that the subspace separation is not an artifact of the specific k chosen.

7 Discussion

7.1 Convergent Evidence for Geometric Separation

Three independent analyses consistently indicate that LoRA adapter feature representations are geometrically separated from base model representations:

1. **Cosine similarity**: Mean max similarity ≈ 0.071 across all 268M comparisons, barely above random (≈ 0).
2. **Principal angles**: Mean ≈ 74 with 0% aligned directions — rigorous subspace-level evidence.
3. **CKA**: Minimum CKA$(h_b, h_\Delta) \approx 0.05$ at layer 18, the key semantic processing layer.

The convergence of three different measures reduces the risk of any single metric being misleading. Cosine similarity could in principle reflect SAE training variance; principal angles address this by operating directly on the full decoder matrices without SAE decomposition; CKA validates at the activation level rather than the weight level.

7.2 The Monitoring Gap

Our findings imply a concrete safety concern: *interpretability tools trained on base model activations may be systematically blind to adapter contributions.* If an organisation deploys a safety-fine-tuned LoRA model and uses Gemma Scope-based analysis to audit it, the SAE's reconstruction failure ($\varepsilon > 1.0$) means the tool is not capturing the adapter's representational contributions. These results suggest that interpretability tools trained solely on base-model activations may incompletely capture adapter-induced representations.

The delta SAE framework we introduce provides a direct solution: training adapter-specific SAEs enables meaningful feature-level auditing of fine-tuned models.

7.3 Rank Effects and Representational Capacity

Rank affects feature density (monotonically increasing at deep layers) and CKA distance (higher rank $\rightarrow$ lower CKA at semantic layers) but does not affect the fundamental geometric novelty ($\sim 74°$ principal angles across all ranks). This suggests that rank controls *how many* weakly aligned features the adapter activates, but not the *nature* of those features — all ranks produce geometrically separated representations.

7.4 Limitations

Single seed: All adapters trained with seed 42; seed stability analysis is deferred to future work.

Single dataset: Results are based on Alpaca instruction tuning; generalisation to other fine-tuning objectives (e.g., domain adaptation, RLHF) requires further study.

Undertrained adapters: Despite loss reduction from $\sim$2.0 to $\sim$1.0, the adapters produce outputs identical to the base model on test prompts, suggesting the base model's strong priors dominate. This precludes causal validation via activation steering and is the primary limitation of the current work.

SAE training scale: Delta SAEs are trained on $\sim$1M token vectors. Larger-scale training may improve reconstruction quality and alter the feature vocabulary.

No baseline comparison: We lack a base-base SAE similarity baseline (two independently trained SAEs on base model activations), which would allow direct comparison to assess whether 0.071 cosine similarity is genuinely low or attributable to SAE training variance. This is a priority for future work.

8 Related Work

LoRA and PEFT: Hu et al. [2022] introduced LoRA; subsequent work explored variants [Dettmers et al., 2024, Zhao et al., 2024]. Hu et al. [2023] studied behavioural effects of fine-tuning, and Biderman et al. [2024] studied forgetting dynamics. None have studied fine-tuning effects at the feature level.

Mechanistic interpretability: SAEs have emerged as a key tool for decomposing superposed representations [Bricken et al., 2023, Templeton et al., 2024]. Cunningham et al. [2023] trained SAEs on GPT-2; Gemma Scope [2024] released comprehensive SAE dictionaries for Gemma-2. All prior work focuses on base models.

Fine-tuning and safety: Qi et al. [2023] and Yang et al. [2023] demonstrated that safety fine-tuning is fragile to subsequent fine-tuning. Evans et al. [2025] showed that narrow task fine-tuning on insecure code causes broad emergent misalignment. Our work provides a mechanistic account at the feature level of why fine-tuning may escape base model safety constraints.

Representation similarity: CKA [Kornblith et al., 2019] and SVCCA [Raghu et al., 2017] have been used to compare model representations across architectures and training runs; we apply these to compare adapter and base model representations within a single model.

9 Conclusion

We presented a systematic mechanistic interpretability analysis of LoRA adapter feature geometry using Sparse Autoencoders. Our delta SAE framework — training SAEs on adapter-induced activation deltas $\mathbf{h}_\Delta = \mathbf{h}_{\text{adapted}} - \mathbf{h}_{\text{base}}$ — provides a mechanistically clean decomposition of adapter contributions.

Three convergent analyses demonstrate that LoRA adapters operate in a feature subspace geometrically separated from the base model: cosine similarity near random (0.071), principal angles consistently near 74°, and $\text{CKA}(h_b, h_\Delta) \approx 0.05$–$0.35$ depending on layer. Adapter-specific SAEs achieve 46–86% lower reconstruction error than base model SAEs on held-out data, confirming the geometric separation reflects genuine learned structure. Feature density scales with rank while geometric novelty remains rank-invariant.

These findings identify a monitoring gap in LoRA-based safety alignment: base-model interpretability tools may be systematically blind to adapter-encoded representations. The delta SAE framework provides a practical tool for feature-level auditing of fine-tuned models.

Future work will pursue causal validation using strongly-differentiated models (gemma-2-9b-it), extend to misalignment detection in the style of Evans et al. [2025], and establish base-base SAE similarity baselines to rigorously bound the observed geometric separation.

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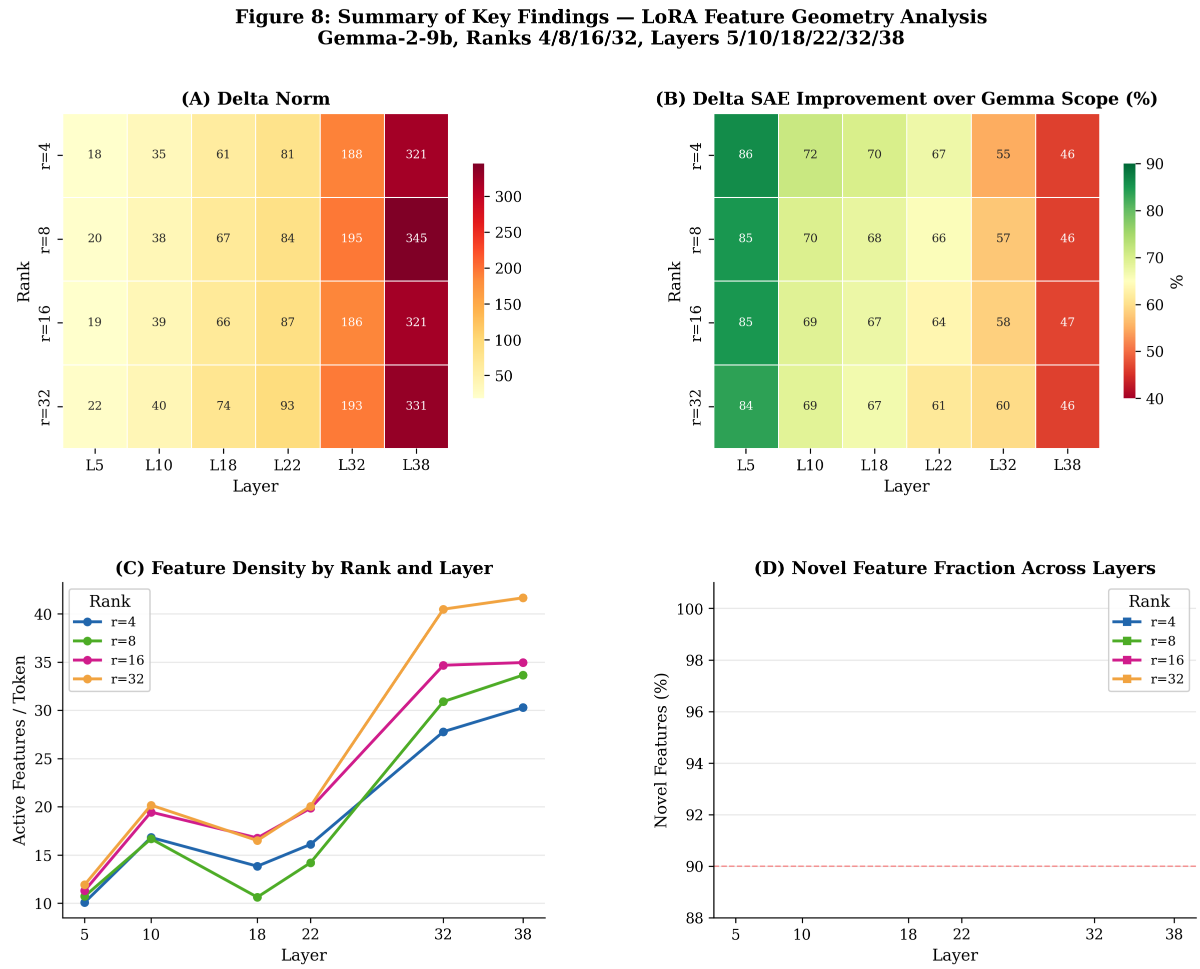

Figure 7: Summary of key findings across all ranks and layers. (A) Delta norm heatmap showing amplification with depth. (B) Reconstruction improvement of delta SAE over Gemma Scope. (C) Feature density scaling with rank and depth. (D) Weakly aligned features fraction — consistently above 93% across all conditions.

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